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ABSTRACT

Introduction: Formulating individualized optimized jump training prescriptions based on the force-velocity profile has become popular, but its effectiveness has been contested. Such training programs have opposite effects on 'maximal average force' and 'maximal average shortening velocity', and we set out to investigate which training-induced changes in the neuromuscular system could cause such effects. **Methods:** We used a musculoskeletal simulation model with four body segments and six muscle-tendon actuators to simulate vertical squat jumps with different loads. Independent input was muscle stimulation over time, which was optimized for maximal jump height. We determined the force-velocity profile for a reference model and investigated how it changed when we modified muscle properties and initial postures. **Results:** We could not reproduce the reported training effects by realistically improving muscle properties (maximal force, shortening velocity and rate of force development) or modifying initial postures of the model. However, the profile was very sensitive to gains in jump height at low-loads but not high loads, or vice versa. Reaching maximal height in force-velocity profile jumps requires skill. We argued that submaximal performance in low-load or high-load jumps caused by lack of skill could be responsible for large imbalances in profiles before training. Differential skill training promoted by the individualized optimized approach could explain quick changes toward a balanced profile. **Conclusions:** If the success of individualized optimized training studies is explained by selective skill improvements, training effects are unlikely to transfer to other tasks, and individualized optimized training will not be superior to other types of training.

Key Words: VERTICAL JUMP HEIGHT, MUSCULOSKELETAL SIMULATION MODEL, PERFORMANCE, OPTIMAL TRAINING

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INTRODUCTION

Vertical jumping ability forms the basis for successful performance in numerous sports. How should a training program aimed at improving jump height be designed? According to Samozino et al. (1) this can best be done on the basis of the force-velocity profile of an individual athlete.

The force-velocity profile may be obtained by having an athlete perform several maximal-effort squat jumps from a fixed starting posture, each with a different external load (Fig. 1). For each jump, the total gain in potential energy from the start of the push-off to the apex of the jump is divided by the lower limb extension range, h_{PO}, to yield what Samozino et al. (2, 3) call the 'average vertical force'; the outcome has the unit of force because J/m=Nm/m=N, but it is formally energy averaged over distance. We will refer to this variable, expressed per kg of body mass, as $\overline{F}^{d}$. $\overline{F}^{d}$ is subsequently combined with $\overline{v}$ of the jump, the average vertical velocity of the center of mass during the push-off, to yield one point in the relation between average force and average velocity. A straight line is fitted to the points of the different jumps, and this line is extrapolated to find $\overline{F}_0^{d}$, the intercept with the $\overline{F}^{d}$-axis, and $\overline{v}_0$, the intercept with the $\overline{v}$-axis. The line is defined by $\overline{F}_0^{d}$ and $\overline{v}_0$, but it can alternatively defined by its slope and $0.25 * \overline{F}_0^{d} * \overline{v}_0$. The latter term is referred to as the maximal average 'power' $\overline{P}_{\max}$ and the slope is referred to as the actual 'force-velocity profile' (1, 2).

How is the force-velocity profile related to jumping performance? In cross-sectional studies it has been established that maximal jump height is strongly correlated to $\overline{P}_{\max}$ and to a lesser extent to h_{PO} (1). Obviously, the same $\overline{P}_{\max}$ can be achieved with different combinations of

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$\bar{F}_0^d$ and $\bar{v}_0$, i.e. with different profiles, and each of these profiles corresponds to a different theoretical maximal unloaded jump height (1, 4). The profile corresponding to the greatest theoretical unloaded jump height is referred to as the optimal profile (Fig. 1), and the extent to which the actual profile deviates from this optimal profile is called the force-velocity imbalance (1, 4). Using multiple regression analysis it has been shown that, in addition to $\bar{P}_{max}$ and h_{PO}, the force-velocity imbalance explains part of the interindividual variation in unloaded jump height: after correction for the effects of variation in $\bar{P}_{max}$ and h_{PO}, athletes with a smaller imbalance tend to jump higher (1).

How is the force-velocity profile used to design an individualized jump training program? Based on the correlations found in cohort studies (1) it has been argued that jump training for a specific athlete should be aimed simultaneously at increasing $\bar{P}_{max}$ and reducing the force-velocity imbalance (1, 4). For that, the athlete may be classified as showing a force deficit (Fig. 1a), showing a velocity deficit (Fig. 1b), or being well-balanced (dashed lines in Fig. 1). An athlete with a force deficit is recommended to try to reduce the imbalance and increase $\bar{P}_{max}$ by force-oriented training, involving exercises at high loads aimed at increasing $\bar{F}_0^d$ (4-7). An athlete with a velocity deficit is recommended to try to reduce the imbalance and increase $\bar{P}_{max}$ by velocity-oriented training, involving high-velocity leg extension exercises aimed at increasing $\bar{v}_0$ (4-6). Finally, a well-balanced athlete is recommended to try and increase $\bar{P}_{max}$ while keeping the profile balanced by training with a combination of exercises aimed at increasing both $\bar{F}_0^d$ and $\bar{v}_0$ (4-6). In this way, jump training is ‘optimized’ for an individual athlete depending on the type and degree of his or her force-velocity imbalance (4-6).

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Is an individualized ‘optimized’ training program based on the force-velocity imbalance more efficacious than a classical, non-individualized jump training program? Jiménez-Reyes et al. (5) were the first to study this. They assigned 84 athletes to three groups: a control group, a group that received classical resistance training for 9 weeks, and a group that received ‘optimized’ training for 9 weeks. The training of an athlete in the ‘optimized’ group depended on the athlete’s force-velocity profile, according to the principles outlined in the previous paragraph. It was found that all athletes in the ‘optimized’ training group increased their jumping performance, whereas the results were much more variable and unclear in the other groups. Furthermore, the increase in unloaded jump height in the ‘optimized’ group was associated with a marked reduction in force-velocity imbalance both for the athletes with a force-deficit and for those with a velocity-deficit. Hence, the authors concluded that an individualized ‘optimized’ training based on the force-velocity imbalance was more efficient for improving jumping performance than a traditional resistance training program. In a later study on 60 similar athletes diagnosed with a force imbalance, Jiménez-Reyes et al. (6) again found that ‘optimized’ training induced unloaded jump height gains of over 10%, associated with large reductions in force-velocity imbalance; the results of that study are shown in Fig. 2. Lindberg et al. (7), however, were unable to replicate these results. They randomly assigned 40 athletes to three groups that trained for 10 weeks. In a group coined ‘toward’, comparable to the ‘optimized’ group of Jiménez-Reyes et al. (5, 6), the training was aimed at reducing the force-velocity imbalance: athletes with a force deficit received force-oriented training and those with a velocity deficit received velocity oriented training. In a second group coined ‘away’, the training was aimed at increasing the force-velocity imbalance: athletes with a force deficit received velocity-oriented training and those with a velocity deficit received force oriented training. In the third group the

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training was balanced, irrespective of the profile of the athletes. Only small gains in unloaded squat jump height were found, on the order of 5%, and the gains of the three groups were not different. Intriguingly, the force-velocity imbalance was not reduced in the 'toward' group, and not increased in the 'away' group. As an example, Fig. 3 shows the training effects in the 'toward' and 'away' group for the force-deficient athletes. This outcome was very different from the outcome of the studies of Jiménez-Reyes et al. (5, 6) (cf. Fig.3a and Fig.2a) and did not support the efficacy of individualized 'optimized' training based on the force-velocity imbalance. Similarly, Lindberg et al. (8) demonstrated that an individualized training intervention based on the force-velocity profile determined using a pneumatic leg press dynamometer was not more effective than a general mixed force- and velocity-based training program in elderly participants.

It is difficult to explain the discrepancy in outcomes of training studies based on the force-velocity imbalance. There are methodological differences among studies in determining the force-velocity profile. For example, Jiménez-Reyes et al. (5, 6) had their athletes perform squat jumps in a Smith machine and calculated takeoff velocity and jump height on the basis of flight time recorded with Optojump (9), whereas Lindberg et al. (7) had their athletes perform free squat jumps and calculated takeoff velocity and jump height from flight time determined from measured ground reaction forces. This has sparked a debate about the accuracy of the different methods (7, 10-13). Surprisingly, however, one important question has not been addressed: What is the explanation for changes in $\bar{F}_0^d$ and $\bar{v}_0$ as a result of training? In the studies by Jiménez-Reyes et al. (5, 6), increases in $\bar{F}_0^d$ as a result of training were accompanied by decreases in $\bar{v}_0$, and increases in $\bar{v}_0$ were accompanied by decreases in $\bar{F}_0^d$ (Fig. 2), with $\bar{P}_{max}$ remaining constant. Hence, the training had opposite effects on $\bar{F}_0^d$ and $\bar{v}_0$, rather than

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concurrent effects. Haug and Payne (14) called such changes ‘profile rotations’, took them for granted, and used a simple model to investigate their effects on vertical jump performance with different constraints. Here, we will not take them for granted but try to identify training-induced changes in the neuromuscular system that may underly them.

In the present study we addressed the question of whether the force-velocity profile for free jumping is a sound basis for individualized jump training prescriptions. We simulated maximal height loaded squat jumps with a musculoskeletal simulation model that has previously been shown to successfully reproduce human vertical squat jumps from various initial postures (15-17) and to have a force-velocity profile similar to that of athletes (18). This allowed us to investigate how changes in muscle properties (maximal force, maximal shortening velocity, and rate of change of force development) and initial posture affect jump height, $\bar{F}_0^d$ and $\bar{v}_0$, and hence the profile. It also allowed us to study how the effects depend on the variables used to determine the force-velocity profile. We specifically tried to understand how increases in $\bar{F}_0^d$ could be accompanied by decreases in $\bar{v}_0$ and vice versa, as observed in the ‘optimized’ training groups of Jiménez-Reyes et al. (see Fig. 2). Based on the outcome of the simulations, we will argue that the training results of Jiménez-Reyes et al. (5, 6) may at least partly be due to differential gains in the athletes’ skill in performing low-load jumps and high-load jumps.

METHODS

The musculoskeletal simulation model (Fig. 4A), which represented a subject with a mass of 82 kg and a body height of 1.92 m, comprised four body segments, actuated by six lumped muscle-tendon complexes of the human lower extremity. Each muscle-tendon complex was a

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Hill-type unit, comprising a contractile element, a series elastic element and a parallel elastic element. Forces of the elastic elements quadratically increased with elongation, while force of the contractile element depended on length and velocity of the contractile element, and active state (19). Active state was related via a first-order process to muscle stimulation *STIM*, a one-dimensional representation of recruitment and firing rate of α-motoneurons (20). *STIM* as a function of time was the only input of the model.

With the musculoskeletal model, we simulated unloaded squat jumps and squat jumps with additional loads at the shoulders in such a way that the total mass of the system ranged between 1.0 and 1.8 times body mass (BM). For each jump, initial *STIM*-levels were found that kept the model in equilibrium in the starting posture. During the jump, *STIM* of each muscle-tendon complex was allowed to linearly increase from the initial level to the maximal level at a rate of 2.2/s; with this value, the model satisfactorily reproduced the average rate of increase of vertical ground reaction force in subjects performing squat jumps (16, their Fig. 5). The increase in *STIM* of a muscle started at an onset time. For each of the unloaded and loaded jumps, the combination of *STIM* onset times that maximised the height achieved by the center of mass was found using a genetic algorithm (21). From the resulting jumps, we determined the force-velocity profile using the equations proposed by Samozino (2, 4, 22).

As reference initial posture we took the posture recommended by Samozino and his group (3, 23). In this posture, the knee angle is 90˚ (full extension being 180˚) and the upper body is almost vertical. Samozino et al. (3, 23) defined h_{PO} as the difference between the height of the greater trochanter in the initial posture and the distance between this trochanter and the

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tips of the toes when the athlete is lying supine with the knees extended and the ankles maximally plantar flexed. They used h_{PO} as an approximation for the vertical displacement of the center of mass during the push-off. Instead of using the greater trochanter, Jiménez-Reyes et al. (5, 6) used the iliac crest; in principle this allows tilting of the pelvis and therewith the trunk to increase h_{PO}, but this does not occur if the trunk is kept upright in the initial posture. In this study, we defined h_{PO} as the increase in height of the center of mass of the system between the initial posture and takeoff in the unloaded jump.

RESULTS

Reference force-velocity profile

Fig. 4b shows the combinations of $\bar{F}_0^d$ and $\bar{v}_0$ for simulated squat jumps from the reference starting posture. In the unloaded jump of the model, h_{PO} was 29.2 cm, jump height was 20.3 cm, and $\bar{P}_{max}$ was 16.6 W/kg. Note that h_{PO}, the vertical displacement of the center of mass during the push-off, was 3.5 cm less than the vertical displacement of the hip because the extension of the legs during the push-off lowers the center of mass of the system within the body (e.g., relative to the greater trochanter). We used h_{PO} in the unloaded jump to calculate all $\bar{F}^d$-values, as proposed by Samozino et al. (2), but it was actually not representative of all jumps. In the loaded jumps h_{PO} was higher (by 2.8 cm in the jump at 1.8 BM); the lowering of the center of mass of the whole system within the body during the push-off is smaller because the relative mass of the legs becomes smaller when more mass is added at the shoulders. We also determined the optimal profile of the model, which turned out to be virtually identical to the actual profile; in the terminology of Jiménez-Reyes et al. (5, 6) the model's actual profile was well-balanced, i.e., it differed less than 10% from the optimal profile.

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Effects of changes in muscle properties

Since the the force-velocity relationship of muscles is the main explanation for why $\bar{F}^d$ is inversely related to $\bar{v}$ (2), we started by determining the effect of increasing the isometric force of all the muscles in the reference model by 20% (Fig. 5A). This caused a gain in unloaded jump height (by 4.7 cm) as well as increases in $\bar{F}_0^d$, $\bar{v}_0$ and $\bar{P}_{max}$. This is because at every shortening velocity, the muscles are producing more force and more power; therefore, they produce more work over their shortening range, so takeoff velocity and hence jump height increase. Next, we determined the effect of increasing the maximal shortening velocity of all the muscles in the reference model by 20% (Fig. 5B). This caused a gain in unloaded jump height (by 3.1 cm), as well as increases in $\bar{F}_0^d$, $\bar{v}_0$ and $\bar{P}_{max}$ (Fig. 5B). The reason is that at every contraction velocity, the relative velocity of the muscles is lower, and according to the force-velocity relation this corresponds to a higher force and power. By changing muscle force and maximal shortening velocity, we could produce a concurrent effect on $\bar{F}_0^d$ and $\bar{v}_0$, but not an opposite effect.

Another variable affecting jump height is the rate of force development (14, 16). This rate of force development has a muscular component, originating from the contraction dynamics of the muscles, and a neural component, which depends on how fast recruitment and firing rate of α-motoneurons are increased by the central nervous system. When we changed the rate of increase of *STIM* in the model from 2.2/s to 5/s, unloaded jump height increased by 2.2 cm, $\bar{F}_0^d$ decreased but $\bar{v}_0$ and $\bar{P}_{max}$ increased (Fig. 5C). This means that increasing the rate of force development affected jump height more in the low-load jumps than in the high-load jumps. This is understandable: in a low-load jump the acceleration is higher than in a high-load jump, and

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therefore the active state, force and work of the muscles are submaximal over a larger part of the push-off distance in the unloaded jump than in a high-load jump (16).

Effects of changes in initial posture

In studies on the force-velocity profile, the initial posture is not well-defined. Typically, it is reported that participants initiated the jumps from a "…comfortable position with a knee angle of about 90˚"(e.g., 5, 6). When we keep the trunk more or less upright, h_{PO} of our simulation model is less than 30 cm and the unloaded jump height seems to suffer substantially from the limited rate of force development (Fig. 5C). Flexing the knees by 10˚ more in the initial posture (knee angle of 80˚ rather than 90˚) increases h_{PO} by about 5 cm and leads to a gain in unloaded jump height by 3 cm, which is in line with the literature showing that jump height increases with push-off distance (2, 4, 15, 24). Furthermore, it has effects similar to increasing the rate of force development: it benefits the work production, but does so less in the high-load jumps than in the low-load jumps; hence $\bar{F}_0^d$ decreases but $\bar{v}_0$ and $\bar{P}_{\max}$ increase (Fig. 5D).

At a given knee angle of 90˚, various initial configurations are still possible; the trunk may be kept upright, but subjects typically prefer to have it tilted forward (e.g., 15, 17). When the model starts from the more natural posture shown in Fig. 6A, h_{PO} is 2.8 cm larger and unloaded jump height 4 cm larger than when the model starts from the reference posture (Fig. 4B). When the trunk is tilted in the initial posture, adding mass at the shoulders causes the center of mass of the whole system to move not only upwards, as in the reference posture (Fig. 4B), but also forwards relative to the hip. At high loads it brings the center of mass of the system as a whole beyond the area of support. In the simulation model, with a fixed knee angle

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of 90˚, we are then forced to make choices. In order to maintain the center of pressure under the foot at the same location as in the unloaded jump, we may rotate the trunk to a more upright orientation as the mass gets bigger, henceforth referred to as 'trunk strategy' (Fig. 6A). Alternatively, we may rotate the whole system around the ankle joint, henceforth referred to as 'ankle strategy'. The two strategies result in different force-velocity profiles (cf. Fig. 6A and Fig. 6B). Obviously, if postural changes occur among loading conditions, it is questionable to use the fixed h_{PO} of the unloaded jump in determining the force-velocity profile. In the case of the 'trunk strategy' (Fig. 6A), the actual vertical displacement of the center of mass of the whole system is 2.7 cm higher in the jump at 1.8 BW than in the unloaded jump, whereas in the case of the 'ankle strategy' (Fig. 6B) it is almost 9 cm higher in the jump at 1.8 BW than in the unloaded jump! This leads to different profiles (cf. Fig. 6A and Fig. 6B).

At this point, a methodological issue is of interest. In order to calculate force-velocity profiles and related variables, some investigators (e.g., 10, 11, 25, 26) have measured the vertical ground force during jumps with a force plate and averaged it over the push-off time. The use of time-averaged vertical force $\bar{F}^t$ yields virtually the same profiles for the 'trunk strategy' (Fig. 6C) and the 'ankle strategy' (Fig. 6D). However, the optimal profiles predict a decrease rather than an increase in theoretical maximal jump height. This can be understood as follows. Fig. 7A shows the vertical ground reaction force as a function of relative push-off time and as a function of relative push-off distance of the unloaded jump. The surface underneath the two curves is different and, hence, $\bar{F}^t$ differs from $\bar{F}^d$, because the displacement per unit time is not constant during the push-off: it starts small and ends large. Given the specific shape of the vertical ground reaction force-time history of the model, $\bar{F}^t$ is less than $\bar{F}^d$. A smaller $\bar{F}^t$ than $\bar{F}^d$

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for each jump leads to a reduction in $\bar{P}_{max}$ (compare Fig. 6C with Fig. 6A, and Fig. 6D with Fig. 6B), which leads to a decrease in theoretical maximal jump height because it outweighs the effects of changes in the slope of the force-velocity relationship (1). The difference between $\bar{F}^t$ and $\bar{F}^d$ depends on the load: in the jump at 1.8 BM it is almost zero (Fig. 7B). This explains why profiles determined using $\bar{F}^t$ differ from profiles determined using $\bar{F}^d$ (compare Fig. 6C with Fig. 6A, and Fig. 6D with Fig. 6B).

Effects of gains in jump height at low-loads but not at high loads, or vice versa

So far, we have studied only maximal height jumps. What if athletes are unfamiliar with certain types of jumps and do not have optimal control, or feel insecure, causing jumps to be submaximal? Without doing further simulations, we may easily investigate the effects of submaximal performance in high-load jumps or low-load jumps. Fig. 8A shows that a 2.5 cm jump height deficit of only the 1.8 BM jump is accompanied by a substantial decrease in $\bar{F}_0^d$ and a substantial increase in $\bar{v}_0$. Conversely, Fig. 8B shows that a 2.5 cm jump height deficit of only the unloaded jump is accompanied by a substantial increase in $\bar{F}_0^d$ and a substantial decrease in $\bar{v}_0$. It follows that if athletes have such jump height deficits at the start of a training program, the profile may substantially be affected by small differential gains in jump height among loading conditions; with differential gains we mean that low-load jump height increases more than high-load jump height, or vice versa.

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DISCUSSION

Formulating individualized training prescriptions based on the force-velocity profile, as proposed by Samozino et al. (1) and implemented in successful training studies (5, 6), has become very popular among trainers and coaches. However, Lindberg et al. (7) could not replicate the success of this training. It has recently been shown that the force-velocity profile does not represent the intrinsic force-velocity relation (18), but that does not necessarily mean that the profile cannot be used as a basis for training interventions. In the present study we simulated maximal height loaded squat jumps with a musculoskeletal model to investigate how changes in muscle properties and initial posture affect jump height, $\bar{F}_0^d$ and $\bar{v}_0$, and hence the force-velocity profile. We specifically wanted to understand which effects of training could explain opposite effects on $\bar{F}_0^d$ and $\bar{v}_0$, as observed in the 'optimized' training group of Jiménez-Reyes et al. (see Fig. 2). Such effects were unexpected, because Samozino et al. (4) claimed that $\bar{F}_0^d$ and $\bar{v}_0$ were independent characteristics of the lower extremity. Below, we will first discuss the insights gained by the outcome of the simulations. We will discuss whether changes in muscle properties may provide the sole explanation for the opposite training effects on $\bar{F}_0^d$ and $\bar{v}_0$ effects. Next, we will discuss the effects of manipulating initial posture of the model, and we will argue that a insufficient standardization of initial posture may have compromised training outcomes in terms of jump height and force-velocity profiles. We will then elaborate on the potential effects of differential performance gains in low-load and high-load jumps. Finally, we will address the question of whether the force-velocity profile for free jumping is a sound basis for individualized jump training prescriptions.

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The force-velocity relationship of muscles is the main explanation for why $\bar{F}^d$ is inversely related to $\bar{v}$ (2, 4). As a result of training, muscle force may increase, and perhaps the maximal shortening velocity of muscles may increase as well (27). Isolated increases of 20% in the maximal force or the maximal shortening velocity of the muscles in the model caused gains in jump height accompanied by increases in both $\bar{F}_0^d$ and $\bar{v}_0$. We have shown that straightforward changes in maximal force (Fig. 5A) or maximal shortening velocity of the muscles (Fig. 5B) are unlikely to cause opposite effects on $\bar{F}_0^d$ and $\bar{v}_0$ as observed in the training studies by Jiménez-Reyes et al. (5, 6). Another variable that may change as a result of training is the rate of force development (28). Increasing solely this rate caused a gain in jump height, accompanied by a decrease in $\bar{F}_0^d$ but an increase $\bar{v}_0$, with an overall gain in $\bar{P}_{max}$ (Fig. 5C). The changes in the force-velocity profile were actually quite similar to the average changes brought about by the training intervention in the velocity-deficient group of Jiménez-Reyes et al. (5, 6) (Fig. 2B). However, we could not think of training-induced changes in muscle properties that could cause an increase in $\bar{F}_0^d$ but a decrease in $\bar{v}_0$, as brought about by the training intervention in the force-deficient group of Jiménez-Reyes et al. (5, 6).

The effects of varying initial postures on jump height and the force-velocity profile (Fig. 5D, Fig. 6) yielded one methodological and two postural caveats. The methodological caveat is that profiles constructed using $\bar{F}^t$, obtained by measuring the vertical ground force and averaging it over push-off time, may be different from profiles constructed using $\bar{F}^d$, obtained by averaging effective work over push-off distance h_{PO} according to the flight-time method (5, 6); this is illustrated in Fig. 7. Since both Lindberg et al. (7) and Jiménez-Reyes et al. (5, 6) used the flight-time method in their training studies, this methodological caveat has no bearing on the

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difference in the outcomes of their training studies (cf. Fig. 2a and Fig. 3a). The other two caveats are related to the initial posture, which is recommended to be a "…comfortable position with a knee angle of about 90˚" (e.g., 5, 6). The first postural caveat concerns the '*about* 90˚': by how much is the knee angle allowed to be different from 90˚? Jiménez-Reyes et al. (5, 6) used the static initial posture before the unloaded jump in determining h_{PO}. They did not report any values, but we could estimate h_{PO} values from individual data provided in (6); they ranged substantially among athletes, from 23.3 cm to 42.2 cm (see Fig. 9A). Considering that the standard deviation in the stature of the athletes was only 5 cm, it is implausible that almost 20 cm variation in h_{PO} is solely due to variation in leg length, so presumably initial knee angles and possibly trunk angles varied as well. Since unloaded jump height of the model could increase by no less than 3 cm after a change in knee angle of only 10˚ (compare Fig. 5C with Fig. 4B), it needs to be ensured in training studies that each individual's knee is not more flexed post-training than pre-training. This may be done, for example, by using a rubber band beneath an athlete's thighs to help the athlete reproduce the initial posture; the height of this band should then be the same in pre-training and post-training measurements (e.g., 7). Jiménez-Reyes et al. (6) did not report taking any precautions. We also estimated post training h_{PO}-values from the individual data provided in (6), and it seemed that they were not very different from the pre-training h_{PO}-values (see Fig. 9A). However, it cannot be ruled out that Jiménez-Reyes et al. (6) determined h_{PO} values only pre-training and that the 'change' in estimated h_{PO} values over the training period is due to variation in the data to which the profile is fitted. The second postural caveat is that at a given knee angle of 90˚, various initial configurations are still possible; the trunk may be kept upright (Fig. 4) or tilted forward (Fig. 6). Starting with a forward-tilted trunk may lead to a gain in jump height of more than 4 cm compared to starting with an upright trunk

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(cf. Fig. 6 and Fig. 4B). However, depending on the strategy chosen ('trunk strategy'or 'ankle strategy') it may elicit large variations in h_{PO} among jumps at different loads that do not warrant the use of the fixed h_{PO} in the unloaded jump in determining the profile, as they cause the profile to be undefined (cf. Figs. 6A and Fig. 6B). Clearly, the statement that the athlete should initiate each jump from a "…comfortable position with a knee angle of about 90°" is not sufficiently prescriptive for a study in which jump height is used for evaluation of the success of a training program that aims to induce beneficial adaptations of the neuromuscular system. Unfortunately, it cannot be decided to what extent changes in initial posture have compromised training outcomes in terms of jump height because none of the training studies referred to in the current paper provided information on pre-training and post-training initial postures.

Let us now turn specifically to the opposite effects of individualized 'optimized' training on $\bar{F}_0^d$ and $\bar{v}_0$. These opposite effects were found in each of the 60 participants of the study of Jiménez-Reyes et al. (6). Fig. 9B shows the individual profiles before and after the training program in that study. We can see that all profiles nicely converge to a balanced profile, but we also see a very wide range of profiles before the start of the training program: profiles varied from -23/s to -4/s! In Lindberg et al. (7, their Fig. 1) there was also a wide range in initial profiles, but they did not all converge to a balanced profile and some diverged away. In the simulation model, it would be impossible to create such a wide range of profiles with a physiologically realistic variation in muscle properties. How can profiles vary so much among athletes, and which training-induced changes in the neuromuscular system can cause them to become balanced in 9-12 weeks? Could this be due to improvements in control, underlying improvements in skill? The theoretical maximal jump height that an athlete can reach in a loaded

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or unloaded jump is determined by the anatomical and physiological properties of that athlete's musculoskeletal system, but the actual jump height reached also depends on control, i.e., the stimulation of each of the muscles as a function of time (29). Optimal control is specific for initial posture (15, 17) and loading conditions (30), and changes in muscle strength require re-optimization of control (29). Hence, in the current study, control was optimized to find the maximal height for each and every jump. Athletes, however, may not have found the optimal muscle stimulation pattern for all the different jumps; the participants in the studies by Jiménez-Reyes et al. (6), for example, were professional futsal or semi-professional soccer and rugby players, and not athletes that regularly jump to maximal height from a squatted posture. Presumably, none of the athletes were familiar with jumping from an initial posture in which the trunk is upright (Fig. 4B), because preferred initial postures for squat jumping typically involve a forward tilted trunk (e.g., 15, 17). Moreover, unloaded jumping is a completely different task than jumping with an additional mass of 80% of body mass at the shoulders. Hence, the athletes' performance of high-load and/or low-load jumps may well have been submaximal to an extent depending on their sports discipline and their experience with the specific jumps used to determine the force-velocity profile, and selective improvement in control may have occurred as a result of training. Imagine a male 'force deficient' athlete FD who performs relatively well at low-load jumps and relatively poor at high-load jumps. His profile will show a force deficit at the start of the training program; a deficit of only 2.5 cm in jump height at 1.8 BM yields a slope of the force-velocity profile of -9.4/s (Fig. 8A). Imagine another male 'velocity deficient' athlete VD who performs relatively poorly at low-load jumps and relatively well at high-load jumps. His profile will show a velocity deficit; a deficit of only 2.5 cm in unloaded jump height yields a slope of the force-velocity profile of -22.4/s (Fig. 8B). Now, FD will engage in strength-oriented

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training, which incorporates for example performing back squat exercises with high loads (5, 6), a motion very similar to a high-load jump. Moreover, the effects on FD's profile will be regularly evaluated (up to every week in (6)), so FD has the opportunity to become less insecure and optimize his motor skill in performing high-load jumps. Perhaps his skill in performing high-load jumps improves more than his skill in performing low-load jumps. Hence, high-load jump height will increase more than low-load jump height, so $\bar{F}_0^d$ will increase and $\bar{v}_0$ will decrease, and the profile will rotate toward the optimal profile (Fig. 8A). At the same time, VD engages in velocity-oriented training, which involves a lot of unloaded jumping exercises (5, 6). Perhaps VD's skill in performing low-load jumps will improve more than that in performing high-load jumps. Hence, low-load jump height will increase more than high-load jump height, so $\bar{F}_0^d$ will increase and $\bar{v}_0$ will decrease, and VD's profile will rotate toward the optimal profile (Fig. 8B). Such changes in skill of performing specific jumps, rather than in muscle properties, may explain why increases in unloaded jump height and reductions in force-velocity imbalance already occur after 3 weeks of training (6). Furthermore, Jiménez-Reyes et al. (6) explicitly tried to reduce each participant's force-velocity imbalance to zero and determine the individual adaptation kinetics. Hence each participant's profile was frequently determined and compared to the optimal profile. Most likely, the participants were aware of the intended outcome of their specific training, and this may introduce significant bias into the results (31). Recently, Lindberg et al. (8) have shown that merely informing participants that they are part of an intervention group causes a placebo training effect on an objective variable such as the 1 repetition maximal back squat performance. This suggests that awareness of being assigned to either a force-oriented or velocity-oriented training program may influence participants' effort levels during testing, potentially skewing the outcomes.

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This brings us to the question of whether the force-velocity profile for free jumping is a sound basis for individualized 'optimized' jump training prescriptions. We have speculated that differential improvements in low-load jumping skill and high-load jumping skill are at least partly responsible for the success of the individualized training program in the studies of Jiménez-Reyes et al. (5, 6). Formally, this is not in contradiction with the theory that $\bar{F}_0^d$ and $\bar{v}_0$ "…characterize the mechanical limits of the entire neuromuscular function, encompassing individual muscle mechanical properties (e.g., intrinsic force-velocity and length–tension relationships, rate of force development), some morphological factors (e.g., cross-sectional area, fascicle length, pennation angle, tendon properties), and neural mechanisms (e.g., motor unit recruitment, firing frequency, motor unit synchronization, intermuscular coordination)" (4). After all, changes in control can be classified as changes in intermuscular coordination. However, low-load jump height will not benefit from an improved intermuscular coordination in high-load jumps, and high-load jump height will not benefit from an improved intermuscular coordination in low-load jumps: the muscle stimulation over time that is optimal for one type of jump is not optimal for another type of jump (15, 17, 29). Although training to become a well-balanced athlete does no harm, balancing the force-velocity profile is not the ultimate purpose of the training; the ultimate purpose is to increase unloaded jump height. If 'force deficient' athlete FD wants to increase his unloaded jump height, it only makes sense for him to engage in strength oriented training if it helps him train something that benefits his unloaded jumping performance. If he lacks skill in performing unloaded jumps, he may as well dedidate his training time specifically to practicing unloaded jumping, as suggested by the result of Lindberg et al. (7) that training 'toward' the optimal profile is not more effective than training 'away' from the profile (Fig. 3). At the same time, the athlete's own sports discipline should not

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be forgotten. As pointed out by Ettema, who called into question the use of force-velocity profiling for designing sprint training interventions (32): “Every training hour used on improving $\bar{v}_0$ is not used on improving $\bar{F}_0^d$, and vice versa.” A trained weight lifter who is not used to performing unloaded jumps may be classified as ‘velocity deficient’ and start a velocity-oriented training program, but also needs to continue performing high-load exercises to prevent a drop in muscle strength and weight lifting performance. Hence, we feel that trainers and coaches should use their own experience-based judgment and not get carried away by the craze for individualized ‘optimized’ jump training prescriptions based on the force-velocity profile.

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FIGURE LEGENDS

Figure 1. Force-velocity profiles for free squat jumps according to Samozino et al. (1). From an initial squatted posture, an athlete performs five maximal height vertical squat jumps. One jump is unloaded, the other four are performed with incremental mass at the shoulders. For each jump, the total change in potential energy from the initial posture to the apex of the jump is divided by push-off distance h_{po} to yield $\bar{F}^d$. $\bar{F}^d$ is then combined with $\bar{v}$, determined as half the takeoff velocity of the center of mass (1), to yield one point in the graph (open circles not on axes, with loading conditions indicated in panel B). A line is fitted to the points obtained from the different jumps, and this line is extrapolated to find $\bar{F}_0^d$ and $\bar{v}_0$, the intercepts with the $\bar{F}^d$-axis and the $\bar{v}$-axis, respectively (open circles on the axes). The slope of the line is called the force-velocity profile. Also, an optimal profile is calculated (1). Compared to the optimal profile, the actual profile may show a force deficit (panel A) or a velocity deficit (panel B). *h*: jump height, i.e. vertical displacement of the center of mass during the airborne phase; $\bar{P}_{max}$: maximal average 'power', equal to $0.25 * \bar{F}_0^d * \bar{v}_0$. BM: body mass (1.0 BM is the unloaded jump, 1.8 BM is the jump with a mass of 0.8 BM at the shoulders). The dot in the stick figures represents the center of mass of the whole system.

Figure 2. Effects of individualized 'optimized' training from the study of Jiménez-Reyes et al. (6). Average results are shown for a group of 28 athletes whose force-velocity profile showed a force deficit (panel A) and for a group of 32 athletes whose force-velocity profile showed a velocity deficit (panel B). Open circles in center of each panel pertain to unloaded jumps.

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Figure 3. Effects of individualized training from the study of Lindberg et al. (7). Average results are shown for two groups of athletes whose force-velocity profile showed a force deficit. One group (n=5) received 'optimized' training 'toward' the optimal profile to reduce the force-velocity imbalance (panel A) and the other group (n=7) received 'non-optimized' training 'away' from the optimal profile to increase the force-velocity imbalance (panel B). Open circles in center of each panel pertain to unloaded jumps.

Figure 4. Musculoskeletal simulation model used in this study (panel A) and its reference force-velocity profile (panel B). The setup of panel B is the same as that of Fig. 1B. The model comprised four body segments actuated by six major muscle tendon complexes of the human lower extremity (SOLeus, GAStrocnemius, monoarticular VASti, RECtus femoris, GLUtei and HAMstrings). Each muscle tendon complex was represented by a Hill type unit, comprising a contractile element, a series elastic element and a parallel elastic element (inset). The 'foot'-segment runs from the ankle joint to the first metatarsophalangeal joint. In the initial posture it is prevented from rotating in a counterclockwise direction by a constraint equation causing the foot to have a 'heel' (grey lines). When the ground reaction moment acting on the foot becomes counterclockwise during the push-off, the constraint equation is released and the foot starts to rotate around the first metatarsophalangeal joint. Additional mass was added at the shoulders (circle on top). The only input of the model was the stimulation of each of the actuators over time, which was optimized to maximize jump height.

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Figure 5. Force-velocity profiles after various manipulations to the reference model. The setup of panels is the same as that of Fig. 1B. Panel A: Maximal force of all muscles was increased by 20%. Panel B: Maximal shortening velocity of all muscles was increased by 20%. Panel C: Rate of change of stimulation, and hence rate of force development of all muscles, was increased. Panel D: Initial knee angle was changed from 90° to 80°.

Figure 6. Force-velocity profiles after manipulations of the initial posture. The setup of panels is the same as that of Fig. 1B. Panels A and B: Profiles created using the method of Samozino et al. (2) for the 'trunk strategy' and 'ankle strategy', respectively. The 'trunk strategy' involves rotating the trunk to a more upright orientation when the mass gets bigger, in order to maintain the center of pressure under the foot at the same location. The 'ankle strategy' involves rotating the whole system around the ankle when the mass gets bigger, with the same purpose. Panels C and D: Profiles created using the time-averaged vertical ground reaction force during the push-off $\bar{F}^t$ instead of the total change in potential energy per unit h_{PO}.

Figure 7. Force during a jump as a function of relative time and relative displacement. The vertical ground reaction force (*F*) during a jump can be averaged over push-off time to yield time-averaged force ($\bar{F}^t$), or averaged over vertical displacement of the center of mass to yield distance-averaged force ($\bar{F}^d$). In the unloaded jump of the model (panel A) the surface under the force curve is bigger when the force is plotted as function of relative distance because the displacement between nodes that are equidistant in time (open circles) is relatively small in the early part of the push-off, and this causes $\bar{F}^d$ to be larger than $\bar{F}^t$. In the jump at 1.8 BM in the 'ankle strategy'(panel B) $\bar{F}^d$ and $\bar{F}^t$ are almost equal, because the initial force is higher.

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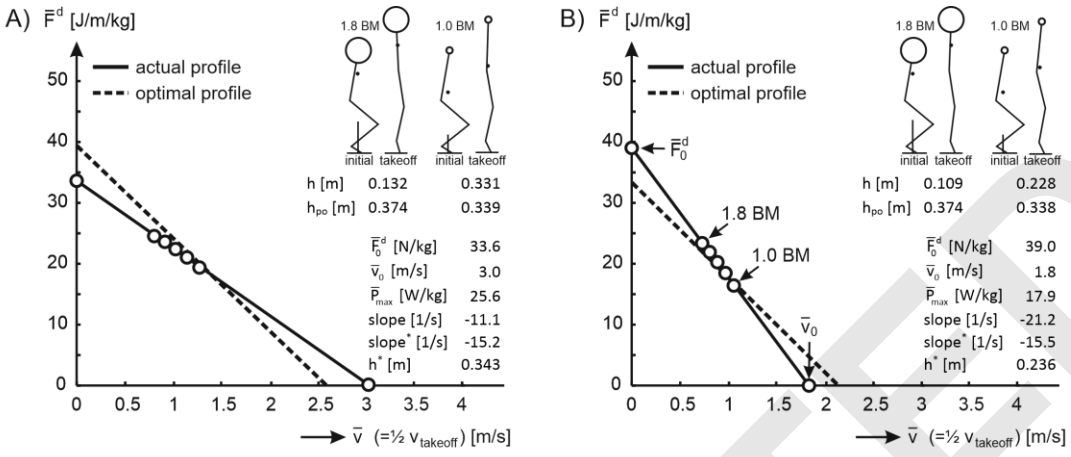
Figure 8. Force-velocity profiles for differential jump height deficits. The setup of panels is the same as that of Fig. 1B, but only the jumps at 1.0 BM and 1.8 MB are represented. We started with the reference profile, with a slope of -16.4/s (Fig. 4B). When we set the height of the jump at 1.8 BM to 2.5 cm below maximal, the slope changed to -9.4/s (panel A). When we set the height of the unloaded jump to 2.5 cm below maximal, the slope changed to -22.4/s (panel B).

Figure 9. Individual effects of individualized 'optimized' training in the study by Jiménez-Reyes et al. (6). Panel A: Values for h_{PO} before and after the training program were estimated from individual data provided in a supplementary file by Jiménez-Reyes et al. (6) using an equation that relates jump height to h_{PO}, $\bar{P}_{max}$ and the force-velocity profile (1). Panel B: force-velocity profiles before and after the training program as documented in the supplementary file by Jiménez-Reyes et al. (6).

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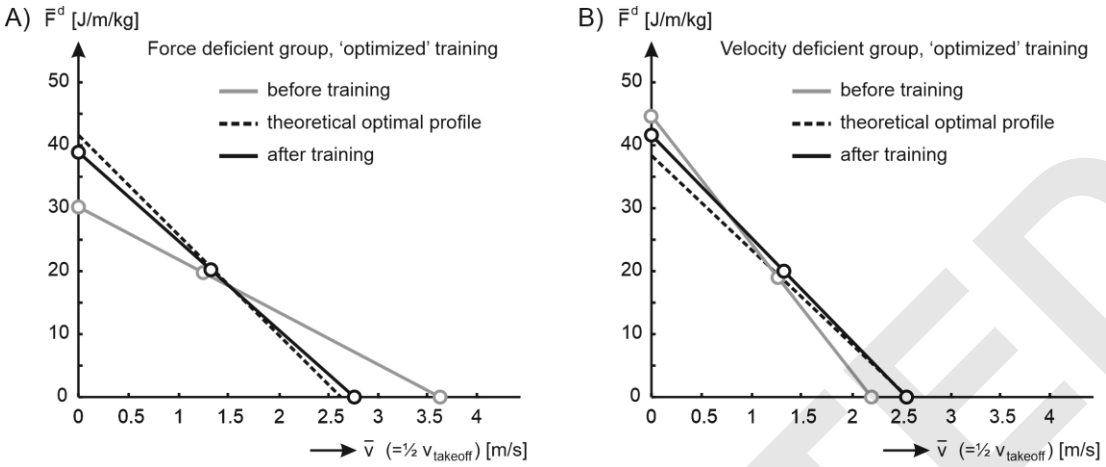
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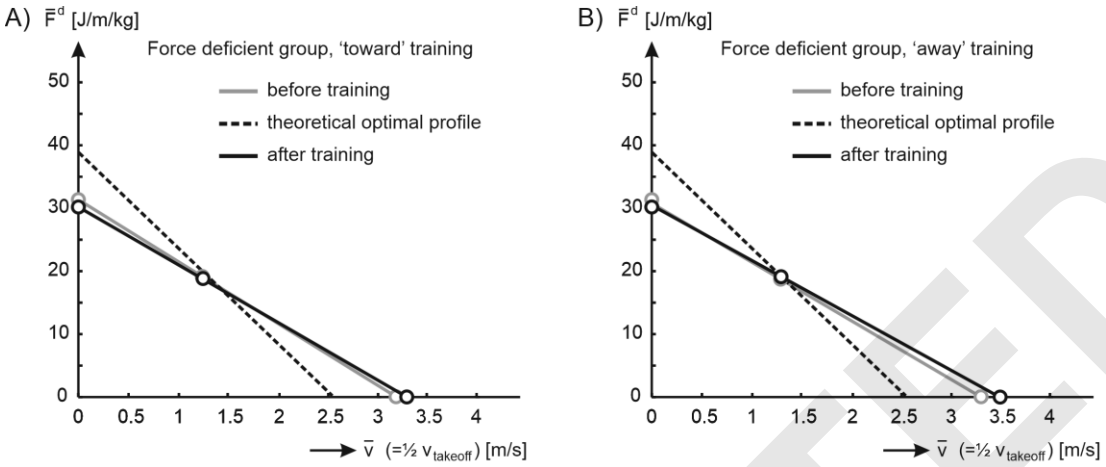
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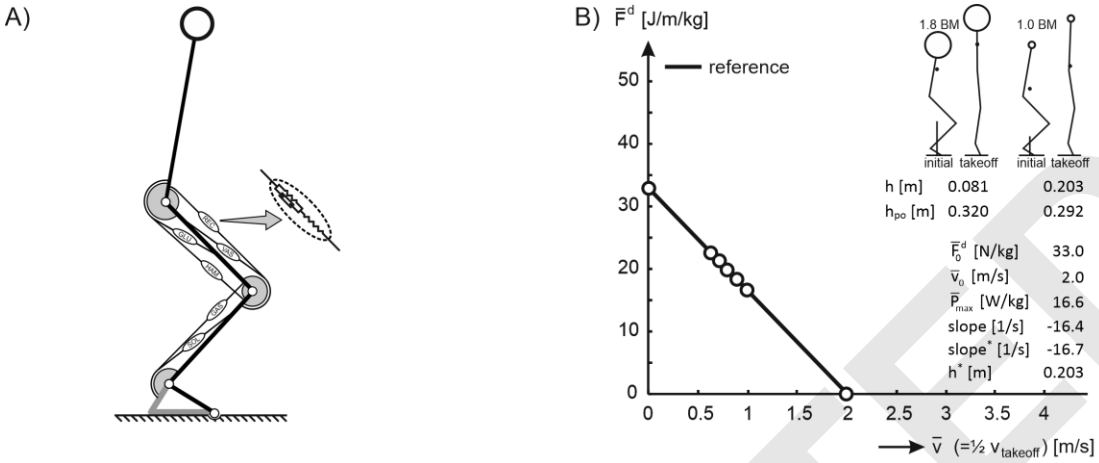
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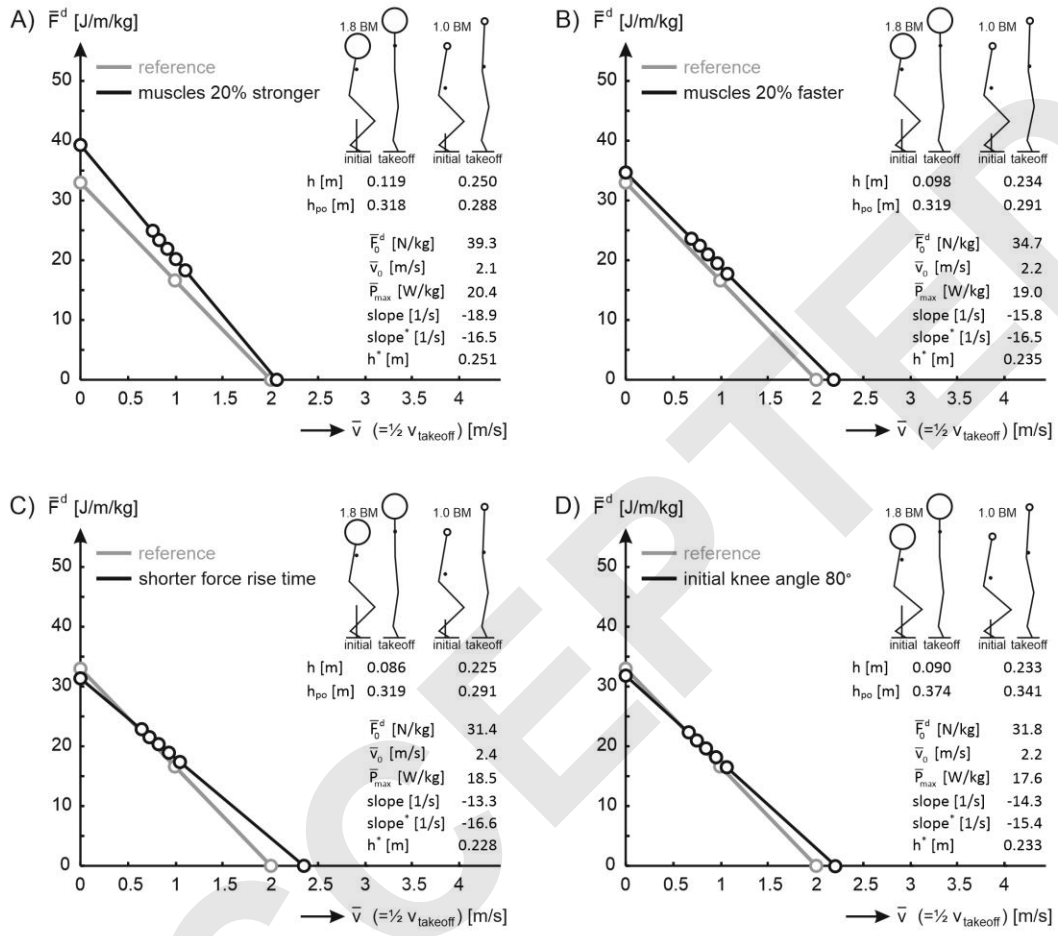
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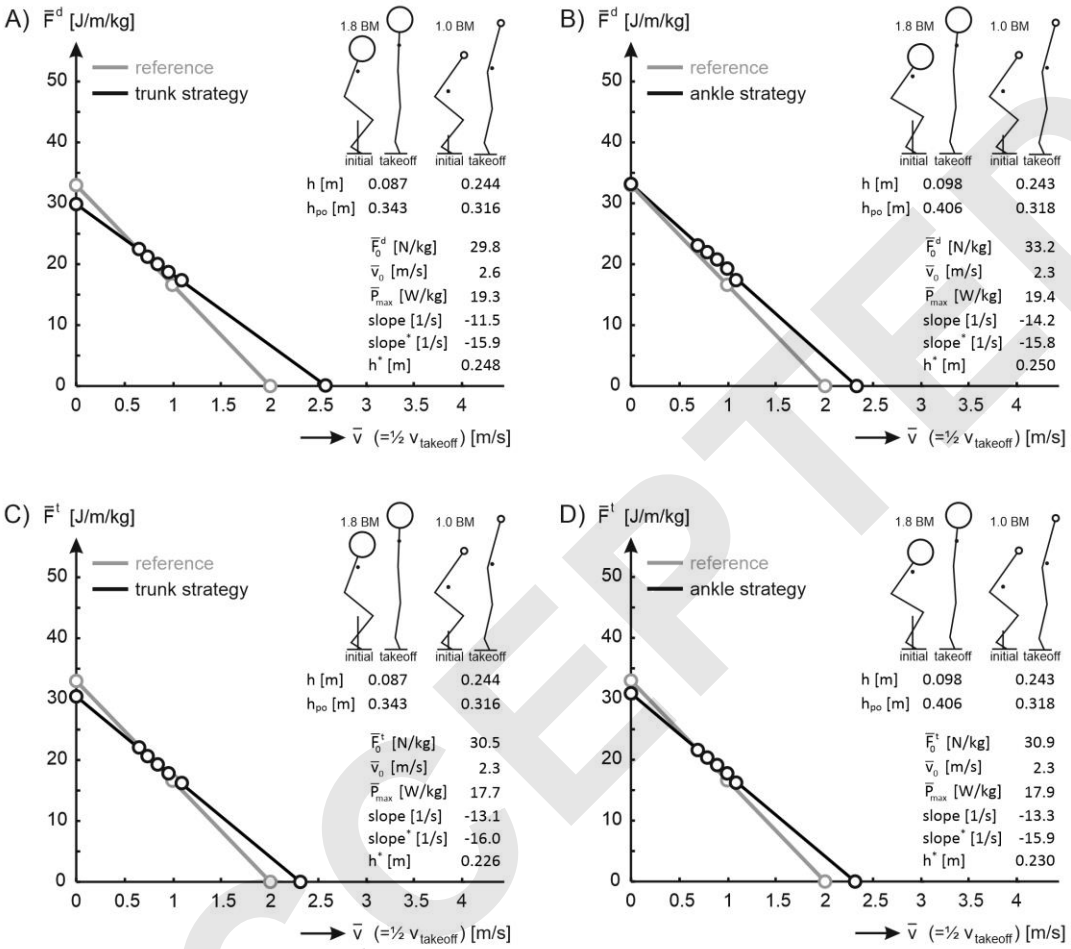
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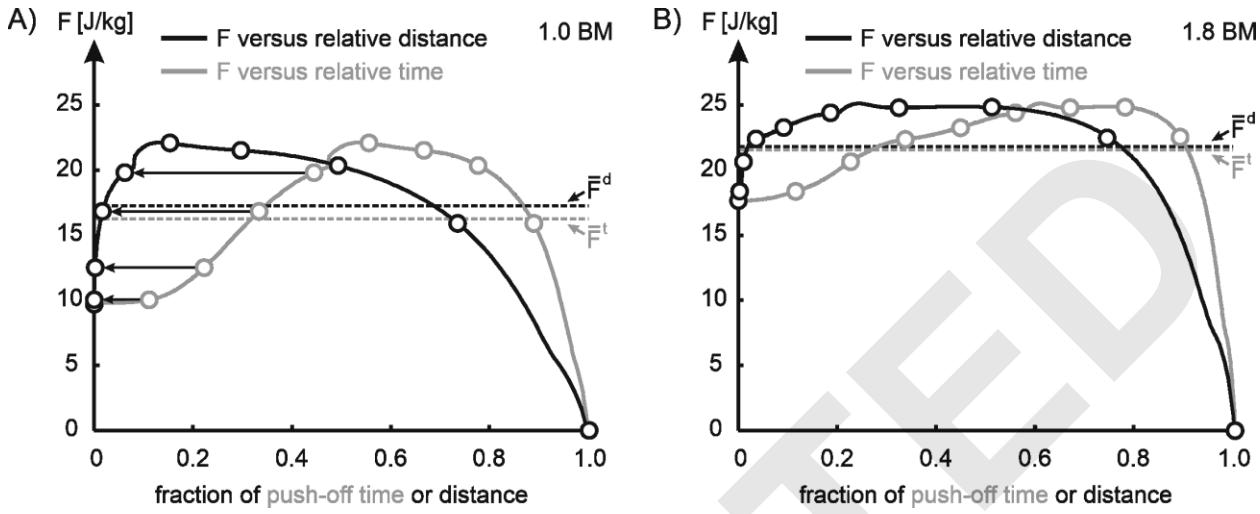
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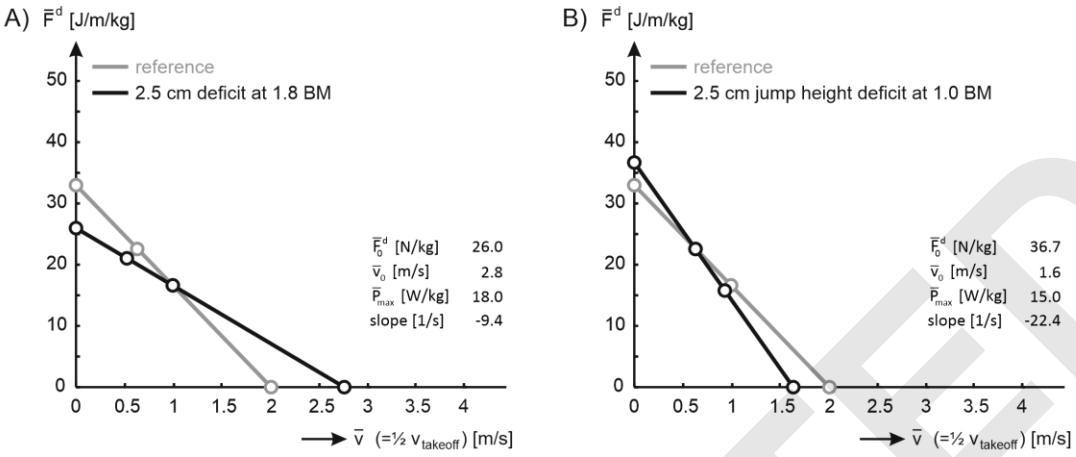
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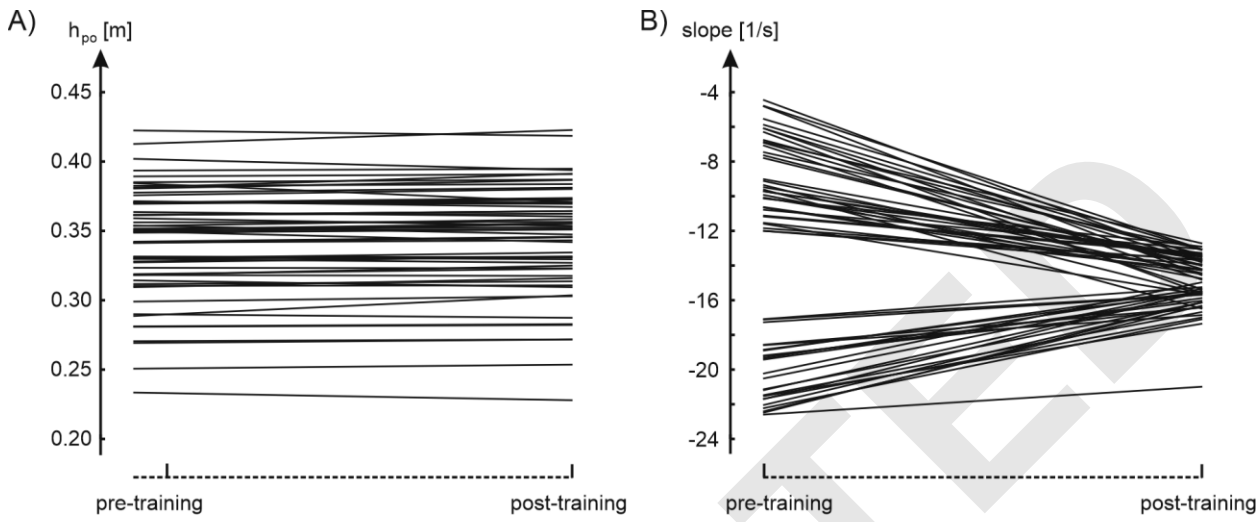
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